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# Bayesian Optimization of Multi-Material 3D-Printed Tensegrity-Inspired Structures for Energy Absorption

*Tensegrity-inspired architectures combine rigid compression members with a continuous network of soft tension elements, yielding lightweight assemblies with tunable nonlinear responses that are attractive for energy-absorbing applications such as protective equipment, packaging, and aerospace landing structures. Multi-material fused-deposition modeling (FDM) of polylactic acid (PLA) struts and thermoplastic polyurethane (TPU) tension elements makes it possible to fabricate diverse tensegrity-inspired geometries rapidly enough to support a high-throughput experimental search, while the rate-dependent damping of TPU is itself an asset for impact mitigation. We report an experiment-driven Bayesian optimization (BO) framework in which a Gaussian-process surrogate is trained directly on physical quasi-static compression and drop-weight impact data and used to recommend the next batch of designs to print and test. The framework parameterizes a family of tensegrity-inspired unit cells over strut geometry, tension-element cross-section, connectivity topology, and unit-cell tiling, and optimizes peak transmitted force, specific energy absorption, and compaction efficiency.*

*Keywords: tensegrity, multi-material 3D printing, Bayesian optimization, energy absorption, design of experiments*

## 1 Introduction

Tensegrity structures are assemblies of rigid compression members suspended within a continuous network of tension members; their defining property, that no two rigid bars touch, yields lightweight assemblies with remarkable strength-to-weight ratios and tunable nonlinear mechanical responses [1,2]. These properties have made tensegrity attractive across scales, from metamaterial unit cells [3–5] to compliant impact-absorbing robots developed for planetary landing [6–10]. Recent work has shown that *tensegrity-inspired* unit cells can be directly 3D-printed and exhibit load-limiting plateaus under drop-weight impact [11], opening a path toward additively manufactured architected absorbers. Throughout, we use *tensegrity-inspired* to denote printable unit cells that adopt the characteristic tensegrity motif—rigid compression members carried by a network of pretensioned soft tension elements—while relaxing the strict classical requirement that no two compression members touch and that the tension members behave as ideal inextensible cables; here the tension elements are extruded TPU and the struts may share printed junctions.

Multi-material fused-deposition modeling (FDM) using polylactic acid (PLA) for rigid struts and thermoplastic polyurethane (TPU) for flexible tension elements enables rapid fabrication of diverse tensegrity-inspired geometries on a single platform [12,13]. Although extruded TPU does not behave as an idealized inextensible cable, its rate-dependent damping is a desirable property for impact-absorption applications and complements the elastic stiffness of the PLA compression members rather than competing with it.

Designing such architected absorbers is fundamentally a *design-of-experiments* problem: the configuration space spanned by strut geometry, tension-element cross-section, connectivity, and unit-cell tiling is large; each candidate must be physically printed and tested; and the relevant performance metrics—peak transmitted force, specific energy absorption (SEA), and compaction efficiency—are noisy and partially correlated. Bayesian optimization (BO) is well-matched to this regime, having driven breakthroughs from hyperparameter tuning of AlphaGo [14] to sustainable concrete [15], autonomous discovery of organic laser emitters [16], and architected materials [17–20].

*1.0.1 Contributions.*. This paper makes the following contributions:

(1) A parameterized family of multi-material 3D-printable tensegrity-inspired unit cells with PLA compression members and TPU tension elements that are anchored *inside* the ends of each PLA strut—the strut acting as a rigid cage in which the cables meeting at a given end join before exiting through discrete outlets—to ensure cyclic interface durability.
(2) An experiment-driven BO loop, built on BoTorch/Ax ~~[21]~~ [21,22], that operates directly on physical quasi-static compression and drop-weight impact measurements, recommending the next designs to fabricate~~without requiring an accurate forward simulator.~~.
(3) ~~A two-fidelity escalation path that pairs the direct-experimental loop above with a planned higher-fidelity rung based on~~ High-fidelity validation experiments using *pretensioned* tensegrity assemblies (true cables and measured pretension) for ~~selected~~ the Pareto-optimal designs ~~; the specific surrogate / fusion strategy is left open.~~ identified by the single-stage loop above, reported separately rather than fused into the optimization.

The remainder of the paper is organized as follows. Section 2 reviews tensegrity mechanics, multi-material 3D printing for energy absorption, and Bayesian optimization with an emphasis on multi-objective and noise-robust acquisition functions. Section 3 introduces the design parameterization, fabrication workflow, experimental protocols, and BO loop. Section 4 reports the results

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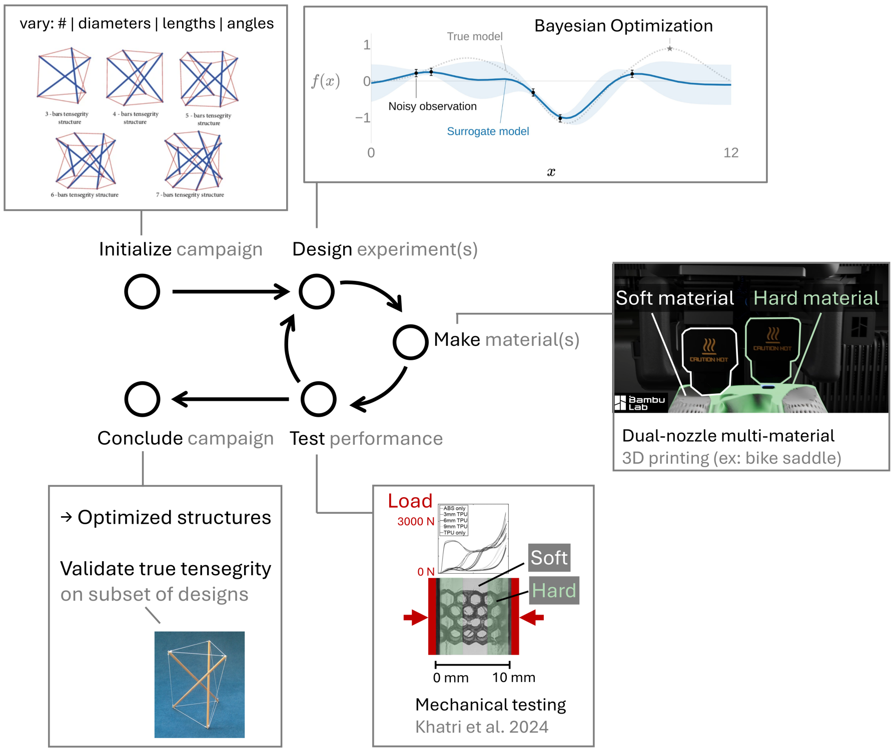


**Fig. 1** Closed-loop, experiment-driven design framework. Parameterized PLA/TPU tensegrity-inspired unit cells are printed, tested under quasi-static compression and drop-weight impact, and the resulting performance data drive a Gaussian-process surrogate that recommends the next batch of designs to fabricate.

of the closed-loop experimental campaign, and Section 5 discusses the implications and limitations of the approach. Section 6 concludes.

## 2 Background

**2.1 Tensegrity-Inspired Architectures for Energy Absorption.** Tensegrity prisms and their planar extensions exhibit softening–stiffening response under compression [3–5], characteristics desirable for impact mitigation because the structure can absorb a large fraction of the input energy at a controlled, sub-injurious force plateau. Santos [23] recently characterized a tensegrity energy-dissipation metamaterial reporting equivalent viscous damping on the order of ~23% with negligible residual strain per impact, underscoring the suitability of tensegrity-inspired architectures for repeated-impact use cases. The space program literature documents large-scale tensegrity-inspired absorbers for planetary landing [6–10] and a long history of crushable energy-absorbing landing systems [24–26]. Pajunen et al. [11] demonstrated that truncated-tetrahedral tensegrity-inspired unit cells can be directly 3D-printed and exhibit load-limiting plateaus, motivating the present focus on additively manufactured TPU/PLA composites. More recent work ~~by Davami et al.~~ on dynamically loaded ~~AM tensegrity unit cells,~~ additively-manufactured energy-absorbing lattices [27], together with Intrigila et al.~~–~~'s bistable tensegrity-like unit cell for lattice metamaterials [28] as the closest published analog~~(SLA Tough 2000, double-T3 quasi-static), confirms that the same~~, confirms that this architecture class also responds favorably under ~~drop-impact~~ impact and quasi-static loading, although neither study employed multi-material FFF or closed-loop design optimization.

**2.2 Multi-Material 3D Printing of PLA/TPU Composites.** Multi-material rigid–soft FDM [12,13] enables PLA–TPU combinations on a single platform with tunable energy absorption. ~~Crucially,~~ Ye et al. [12] introduced a *wrapping-based* strategy in which rigid cores are encapsulated by continuous soft skins, preventing interface delamination and enabling cyclic ~~durability—a critical enabler for our fabrication approach~~durability, something we take inspiration from in our designs. Recent multi-material PLA/TPU sandwich and layered composites [29,30] report quantitative bounds on stiffness, strength, and energy absorption for the same PLA–TPU pair used here, while ~~FFF PLA~~ fused filament fabrication (FFF) PLA fatigue data [31,32] inform conservative design bounds for the rigid struts. Architected multi-material lattices with co-printed compliant skins have been explicitly designed for high specific energy absorption [33], with FFF-specific multimaterial interface adhesion (notably TPU bonded to rigid co-printed substrates) characterized in [34]; recent surveys catalogue the rapidly expanding additively manufactured polymeric energy-absorption literature [35,36].

**2.3 Bayesian Optimization for Architected-Material Design.** Bayesian optimization is a sequential, model-based strategy for optimizing expensive black-box functions using a Gaussian-process surrogate [37,38]. Modern implementations such as BoTorch [21] and the Ax platform built atop it [22] support parallel batched evaluations and a wide variety of acquisition functions; recent algorithmic advances include log-EI for numerical robustness [39], noisy expected hypervolume improvement for multi-objective settings [40], input-noise-robust acquisitions [41], and evolution-guided constrained multi-objective BO for self-driving labs [42]. Mixed/categorical search spaces—needed here for the discrete connectivity-topology variable—have dedicated treatments in BOCS [43] and the one-hot-with-rounded-kernel construction of [44]. BO has been applied to mechanical metamaterial and structural design [17–19] and to additive manufacturing [20]. Mo et al. [17] in particular established a multifidelity BO framework for architected materials by fusing low-fidelity simulations with high-fidelity ground truth via nonlinear information-fusion priors [45]; the present work shifts the emphasis from simulation surrogates to direct experimental data, appropriate for materials whose constitutive response is difficult to calibrate from first principles.

## 3 Materials and Methods

**3.1 Design Parameterization.** We parameterize a family of tensegrity-inspired unit cells via four groups of design variables:

- **PLA compression members.** Strut diameter $d_s$ and length $\ell_s$, with the slenderness ratio $\ell_s/d_s$ bounded away from buckling-dominated regimes.
- **TPU tension elements.** Cross-sectional area $A_t$ (or equivalent diameter $d_t$ for circular cross-sections) and pre-tension geometry imposed during printing.
- **Connectivity topology.** Selected from a discrete set of candidate unit cells (e.g., truncated-tetrahedral, prism-based, and octahedral variants); the design vector encodes a categorical choice.
- **Unit-cell tiling.** Number of cells $N_x \times N_y \times N_z$ and the orientation of each cell relative to the impact direction.

*3.1.1 Working prototype~~.~~* The first instance of this family used to validate the fabrication and test pipeline is a three-strut ($T_3$) tensegrity prism scaled to a ~50 mm bounding box, with $d_s$ = and a cable diameter $d_t$ swept over the FFF-resolvable set $\{1.2, 1.8, 2.4, 3.0, 4.5\}$ mm. Under the prism's $D_3$ symmetry the twelve member-diameter axes collapse to four orbit axes (one strut orbit and three cable orbits: saddle, top, bottom), which keeps the initial design vector low-dimensional while leaving room to relax to per-member diameters once the saddle/top/bottom orbits have been individually characterized.

**Table 1** Design variables for the $T_3$-prism Sobol initialization batch, as implemented in the BO harness (single-iteration $n = 9$ Sobol set). The joint diameter is held fixed at the `t3-prism.scad` default; all specimens are printed in the vertical orientation and packed $3 \times 3$ on a single Bambu Lab H2D plate.

| Design variable | Symbol | Units | Range |
|---|---|---|---|
| Base radius | $R$ | mm | 25–40 |
| Cell height | $H$ | mm | 60–110 |
| Twist angle | $\theta$ | deg | 40–80 |
| Strut (PLA) diameter | $d_s$ | mm | 6.0–12.0 |
| Cable (TPU) diameter | $d_t$ | mm | 3.0–5.5 |
| Joint diameter (fixed) | $d_j$ | mm | 7.0 |

*Figure placeholder: Two printed $T_3$-prism prototypes with callouts labeling the design variables of Table 1 (strut diameter $d_s$, cable diameter $d_t$, base radius $R$, height $H$, and twist angle $\theta$); pairs with the design-variable table.* Use actual printed-specimen photographs from the PR #35 print history.

**Fig. 2** ***Placeholder.*** **Two printed $T_3$-prism prototypes with callouts labeling the design variables of Table 1 (strut diameter $d_s$, cable diameter $d_t$, base radius $R$, height $H$, and twist angle $\theta$); pairs with the design-variable table. *Use actual printed-specimen photographs from the PR #35 print history.***

**3.2 Multi-Material Fabrication.** Specimens are fabricated on a multi-material FDM printer using PLA for the rigid struts and TPU for the soft tension elements. Rather than wrapping the struts in a continuous TPU skin, the current design anchors the TPU tension elements *inside* the ends of each PLA strut: the cables converging at a given strut end are joined within the strut, which acts as a rigid cage housing the junction, and the individual tendons then exit through discrete outlets. This keeps the soft–rigid interface in internal, compression-dominated pockets rather than at an exposed overmolded surface. This construction effectively inverts the material roles of Ye et al.'s soft-skin-over-rigid-core wrapping [12] : here a soft TPU core is captured within a rigid PLA shell, so we cite that work only as a multi-material co-printing precedent rather than as validation of the present junction.

*3.2.1 Print platform and joint geometry.*. The working platform is a Bambu Lab H2D printer, which permits a single rigid–soft build without filament swaps. Strut endpoints are tied to cables through parametric joint geometries developed and ranked through a five-design OpenSCAD study (anchor-bulb, dovetail, TPU-sleeve overmold, eyelet-loop, and TPU-rebar variants); the working prototype uses a dovetail joint (Design B) with an anchor-bulb backup (Design A) for sensitivity studies, with a captive TPU core routed inside the PLA shell to keep the soft tendon protected from layer-line failure at the strut-to-cable transition. The full five-design comparison and the selected junction are detailed in the Supplementary Information.

*3.2.2 Supports for soft members.*. Because near-vertical TPU tendons are otherwise unsupported during printing, the slicer profile combines a tensegrity-specific Bambu Studio recipe (support threshold angle dropped from 40° to 10°, support material matched to the rigid extruder) with manually generated narrowing pillars that ray-cast against the printable mesh underside, producing TPU-safe coverage of vertical members without slicer-side painting. Full workflow details are provided in the Supplementary Information.

**Table 2** Multi-material print parameters on the Bambu Lab H2D. Cells marked *TBD* are pending confirmation; the reporting structure follows the methods section of Ye et al. [12] .

| Parameter | PLA (rigid struts) | TPU (soft tendons) |
|---|---|---|
| Nozzle temperature (°C) | *TBD* | *TBD* |
| Bed temperature (°C) | *TBD* | |
| Layer height (mm) | *TBD* | |
| Print speed (mm/min) | *TBD* | *TBD* |
| Infill (%) | *TBD* | *TBD* |
| Build-plate type | Textured PEI | |
| Build orientation | Vertical | |
| Supports | Off in slicer; manual narrowing pillars | |

**3.3 Experimental Characterization.** Each printed specimen is subjected to two physical tests:

*3.3.1 Quasi-static compression.*. A uniaxial load–displacement curve is recorded on a benchtop electromechanical load frame (BYU CB 123 Instron) following best-practice combinations of ASTM D638, ASTM D412, ASTM E111, ASTM F2971, and ISO/ASTM 52900/52921 adapted to architected polymer cells. Because expected cell stiffnesses fall in the ~5–30 g force range, the default 5 kN load cell is over-ranged and is being replaced by a 100–500 N cell for the first campaign.

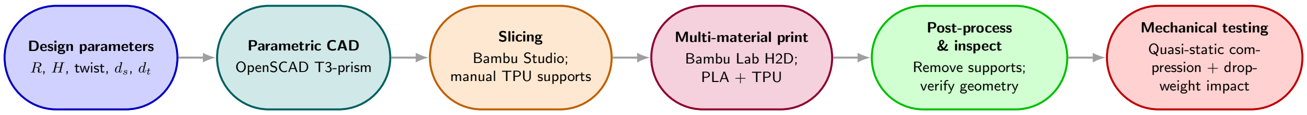


**Fig. 3** Fabrication and characterization workflow for the multi-material tensegrity-inspired unit cells: design parameters drive parametric CAD, slicing with manually generated TPU supports, a single multi-material print on the Bambu Lab H2D, post-processing and inspection, and mechanical testing.

Reported metrics are

$$\mathrm{SEA} = \frac{1}{m}\int_0^{\delta_d} F(\delta)\,\mathrm{d}\delta, \qquad \eta_c = \frac{\int_0^{\delta_d} F(\delta)\,\mathrm{d}\delta}{F_{\max}\,\delta_d}, \tag{1}$$

where $m$ is the specimen mass, $\delta_d$ is the densification displacement, $F_{\max}$ is the peak transmitted force, and $\eta_c$ is the compaction efficiency.

*3.3.2 Drop-weight impact.*. Following Pajunen et al. [11], specimens are subjected to drop-weight impact tests with instrumented load cells; we report the peak transmitted force and the energy-absorption plateau characteristics. The primary fixture is a bungee-assisted laboratory drop tower in which the base accelerates downward faster than $1g$; because unconstrained specimens lift off the base during descent by design of the rig, the protocol constrains the specimen top through light tethers (capping upward travel) and registers the base via transfer tape or V-block features so that loading-direction compliance is not altered. A higher-fidelity replicate campaign on a Lansmont M23 drop tower with synchronized Polytec QTec laser vibrometry is planned for selected Pareto-optimal designs. Slip resistance and traction at the specimen–floor interface—governed by the tip geometry rather than the internal lattice—follow the test methodology of ISO 11334-4 [46] for walking-aid tips and are out of scope for the present axial-impact study.

*3.3.3 Per-modality objectives.*. Five complementary data streams (triaxial accelerometer, high-speed video, shaker transfer function, slug-firing gas gun, and Polytec QTec laser-Doppler vibrometer) each contribute a distinct raw observable; the mapping from each stream to BO-ready scalar objectives (SEA, VEA, $F_{\max}$, cushion-curve intercept, Gibson–Ashby relative density) is defined in five companion briefs and consolidated in a cross-modality synthesis.

**3.4 Experiment-Driven Bayesian Optimization Loop.** The optimization loop maintains a Gaussian-process (GP) surrogate over the design space defined in Section 3.1. After each round of physical experiments, the GP is updated and an acquisition function is maximized to recommend the next batch of designs to fabricate. Implementation uses BoTorch [21] with the log-Expected-Improvement (LogEI) acquisition [39] for the single-objective baseline and noisy Expected Hypervolume Improvement (qNEHVI) [40] for the multi-objective case (simultaneously maximizing SEA and compaction efficiency while constraining peak transmitted force). When experimental noise dominates (e.g., between-print variability), we additionally evaluate the input-noise-robust acquisition of Daulton et al. [41] and the evolution-guided constrained variant of Low et al. [42]. The discrete connectivity-topology variable is handled with the mixed-search-space construction of [44]; the BOCS combinatorial baseline [43] is reported for comparison.

*3.4.1 Implementation status.*. The campaign harness is built on Ax's `HierarchicalSearchSpace` so that printer-feasibility constraints (e.g. the cable-diameter categorical depending on the chosen connectivity topology) can be encoded directly in the search space rather than enforced post hoc. The first physical batch consists of $n = 9$ specimens generated by a Sobol sequence over the four $D_3$-orbit axes of the working $T_3$ prism and printed in a single $3\times3$ plate layout. For the present ≤25-dimensional parameterization the default acquisition pairing is sparse axis-aligned subspace BO (SAASBO) with qNEHVI; escalation to trust-region BO (TuRBO) is planned only if larger per-member-diameter tilings push the design space well beyond this regime.

*Figure placeholder: Leave-one-out cross-validation of the Gaussian-process surrogate (predicted vs. measured SEA and peak transmitted force), once campaign data are available.*

**Fig. 4** ***Placeholder.*** **Leave-one-out cross-validation of the Gaussian-process surrogate (predicted vs. measured SEA and peak transmitted force), once campaign data are available.**

*Figure placeholder: Parameter-sensitivity ranking (e.g., SAASBO inverse length-scales or Sobol indices) of the design variables in Table 1.*

**Fig. 5** ***Placeholder.*** **Parameter-sensitivity ranking (e.g., SAASBO inverse length-scales or Sobol indices) of the design variables in Table 1.**

# 4 Results

## 4.1 Convergence of the BO Loop.

*Figure placeholder: Best-so-far SEA (and Pareto hypervolume for the multi-objective runs) vs. number of physical experiments, averaged over random seeds; baselines: LHS and random search.*

**Fig. 6** ***Placeholder.*** **Best-so-far SEA (and Pareto hypervolume for the multi-objective runs) vs. number of physical experiments, averaged over random seeds; baselines: LHS and random search.**

### 4.2 Pareto-Optimal Designs.

*Figure placeholder: Pareto front in (peak transmitted force, SEA) space; representative geometries annotated.*

**Fig. 7** ***Placeholder.*** **Pareto front in (peak transmitted force, SEA) space; representative geometries annotated.**

### 4.3 Reproducibility Across Print Replicates.

**Table 3** ***Placeholder.*** **Mean and CV of SEA, peak force, and compaction efficiency across *n* replicate prints for the top-*k* designs.**

| Column A | Column B | Column C |
|---|---|---|
| | *(table contents pending)* | |

## 5 Discussion

Topics to address:

- Comparison to the simulation-driven multifidelity framework of Mo et al. [17] and the single-condition fabrication study of Pajunen et al. [11].
- Practical considerations: TPU–PLA interface durability under cyclic loading; sensitivity of optimal designs to print parameters (extrusion temperature, layer height); transferability to TPU/PETG or other high-temperature thermoplastics.
- Path forward toward integrated tread+lattice structures, miniaturization to crutch-tip envelopes ( from the Edison crutch-tip synthesis), and field testing. Long-term crutch use is associated with substantial upper-extremity overuse injury and ground-reaction-force impulses [47–51]; existing spring-loaded [52,53] and polymer-damped [54] tip designs reduce loading rate but not peak transmitted force, leaving an open design opportunity that an architected tensegrity-inspired insert is well-positioned to address. Recent open-source 3D-printed crutch work [55] and expert evaluations of orthopaedic caps [56] suggest a plausible translation path; adjacent footwear and orthotic lattice studies [57–59] provide additional transferable design heuristics.

#### *5.0.1 Limitations..*

## 6 Conclusions

We have presented an experiment-driven Bayesian-optimization framework for designing multi-material 3D-printed tensegrity-inspired energy absorbers using PLA struts and TPU tension elements. The framework parameterizes a family of unit cells, fabricates each candidate on a single multi-material FDM platform with TPU tendons anchored inside the PLA strut ends, characterizes the response under quasi-static compression and drop-weight impact, and updates a Gaussian-process surrogate [21] that recommends the next batch of designs. The approach is broadly applicable to architected absorbers whose constitutive response is difficult to calibrate from first principles and where physical experiments—rather than simulations—are the natural ground truth.

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## Conflict of Interest

The authors declare no conflict of interest.

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## List of Figures



## List of Tables